

Direct detection and classification of *E. coli* and fecal streptococci using an optical water droplet method and convolutional neural networks

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Detecting harmful bacteria in drinking water is a significant concern for public health. Indicator bacteria like *E. coli* and fecal streptococci serve as markers for fecal contamination of water [Environmental and Pollution Science (2019), pp. 191]. Several detection methods of these markers require costly equipment, and specialized laboratories and technicians, and are time-consuming, resulting in labor-intensive processes [J. Phys. Conf. Ser. 995, 012065 (2018)]. This paper proposes an optimal method that combines an optical water droplet method and convolutional neural networks (CNNs) to provide an accurate and cost- and time-effective approach to directly detect and classify *E. coli* and fecal streptococci in water. The system captures images of indicator bacteria and then classifies them. We obtain a classification accuracy of up to 0.89 and a loss of 0.13. This work constitutes a step toward an integrated, real-time, and automatic optical-based detection system for water-borne pathogenic agents. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Due to the shortage of potable water, only 24% of Sub-Saharan Africa has access to potable water [1]. Moreover, about 2.5 billion people worldwide suffer from a lack of quality drinking water, especially in developing countries [2]. Drinking contaminated water can be extremely dangerous and may lead to water-borne diseases [3]. For example, diarrhea, one of these water-borne diseases, leads to the death of over 800,000 children annually [4].

To know if water is contaminated, one needs to check if it contains indicator bacteria such as *Escherichia coli* (*E. coli*) and fecal streptococci [5]. *Escherichia coli* (*E. coli*) is a rod-like shape and is one of the major causative bacteria for water-borne diseases [6]. Alternatively, fecal streptococci refer to a collection of gram-positive, spherical bacteriophages and are part of the coliform bacteria group. The source of these indicator bacteria is usually the feces or other wastes of humans and various warm-blooded animals [7]. These indicator bacteria often do not cause illnesses directly but have demonstrated characteristics that make them good indicators of harmful microorganisms present in water such as *vibrio cholerae* and *salmonella typhi* [7,8].

It is then crucial to prioritize their detection. Conventional techniques such as multiple tube fermentation, membrane filtration, culture and colony counting, mass spectrometry, and

more are implemented [9]. However, they present some drawbacks such as a long total analysis time, viable but non-culturable (VBNC) bacteria, and requiring huge equipment [9].

Compared to the above, optical methods offer some advantages: rapid, non-invasive, and amplification-free measurements [10]. These optical methods include colorimetry, fluorescence with bio-markers, CMOS-based imagery with holography, among others [4,11–14]. Despite these advantages, these optical methods are still time-consuming and expensive, and the sample preparation is sometimes complex making them error-prone [13].

It is necessary to explore the optical water droplet method [15]. It constitutes a low-cost microbe magnification technique whose image can be easily captured for processing. In addition, combining the optical water droplet method with image classification techniques like convolutional neural networks (CNNs) would bring a general benefit in terms of time, equipment, and process. Our interest is focused on convolutional neural networks (CNNs) because they have been successfully used for bacteria detection and are adapted to the context where a dataset is concerned [16–18].

The rest of the paper is organized as follows. Section 2 briefly presents the optical water droplet method, the acquisition of

water sample images, and their classification. In Section 3, simulation results are presented and discussed. Then, the last section is the conclusion.

2. MATERIALS AND METHODS

A. Water Sample

For this work, water samples are collected in the Dschang municipality, Cameroon, in water stations that were proven to permanently and abundantly contain *E. coli* and fecal streptococci [19–21]. These water stations are: (i) a wastewater treatment station located between campuses A and G of the University of Dschang [19,20]. It collects wastewater from the students' residence for treatment and has a pretreatment section, a digester chamber, and accumulation and infiltration chambers; (ii) the water collection points of Madagascar in the Dschang municipality [21].

B. Detection Method

The proposed process of detecting and classifying *E. coli* and fecal streptococci in the water is done by using the flowchart in Fig. 1. Each step of the flowchart is developed hereafter.

1. Image Acquisition

The first step in this process is the image acquisition. The experimental setup (Fig. 2) is inspired by the optical water droplet method proposed by Planistic [15]. This optical water droplet method is proven to display important magnification. Here, equipment like a laser, a water sample, sterile laboratory glass bottles, and a sterilized syringe are used [Fig. 2(a)]. A syringe whose piston is pushed to form a water-drop at the end of the nozzle is positioned at a certain distance d from a screen. A green laser beam of wavelength 532 nm pointed through the median of the water-drop perpendicular to the screen magnifies the shadow images of several bacteria floating and moving [Fig. 2(a)]. The green laser was used because it has less penetration in organism tissues and then can give a better shadow of the

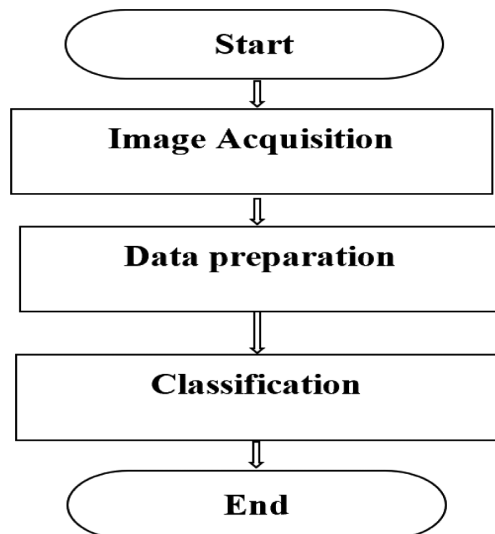


Fig. 1. Flowchart of the proposed optical-based detection of bacteria in water.

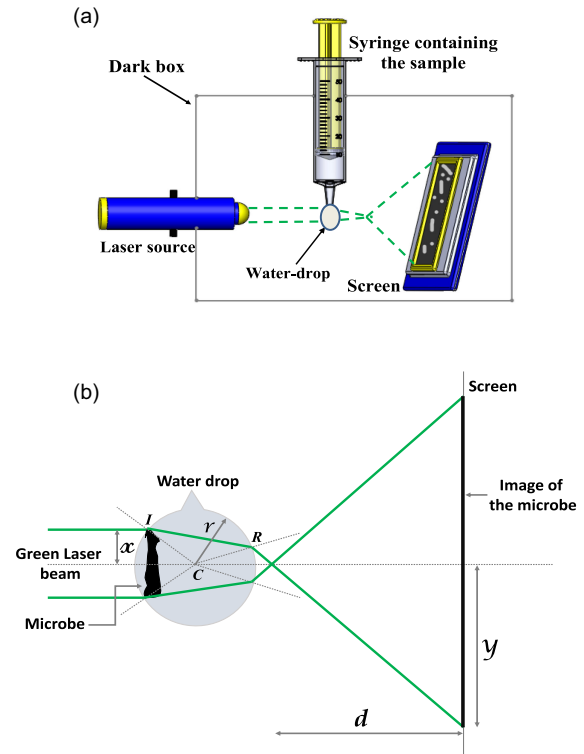


Fig. 2. (a) 3D model of the experimental setup. One can identify the laser beam, the water-drop, and the screen. (b) The laser beam trajectory through the water-drop and the microbe magnification on the screen (inspired from [15]). In (b), the dashed lines are the axis. C represents the center of the water-drop. I and R are the incident points on the input and output surfaces of the water-drop, respectively.

bacteria [22]. The setup is placed in a closed dark box to avoid water contamination from microorganisms floating in the air and perturbation from ambient light.

For mathematical convenience and without loss of generality, the water-drop at the end of the syringe is taken as spherical with center C [Fig. 2(b)]. Microbes of size $2x$ in the water-drop of radius r behave like an opaque medium for the green light beam that flows through the water-drop. That light beam refracts twice at I and R as it passes through the water-air interfaces. The emerging rays reach the screen at a distance d on which the microbe shadow size is $2y$. Using Snell-Descartes laws in paraxial approximation, it was successfully demonstrated that the magnification of the microbe $M = \frac{y}{x}$ is defined as [15]

$$M = \frac{d}{r} \left[\frac{1}{u} \tan \left(2 \left(\arcsin u - \arcsin \frac{u}{n} \right) \right) \right], \quad (1)$$

where n is the refractive index of the water. $u = x/r$ is the fraction of the microbe size to the water-drop radius, thereafter called the relative microbe size. Then, u is taken within $]0, 1[$. The magnification is proportional to the distance d between the water-drop and the screen [15]. Figure 3 shows the plot of the magnification as a function of the relative microbe size for different values of the distance d . For better analysis, the recommended magnification for bacteria is between $400\times$ and $1000\times$ [23]. Such condition is satisfied for d within 1 m and 2 m with a water-drop of radius $r = 3$ mm and refractive index equal to 1.33 (Fig. 3). The magnification slightly grows

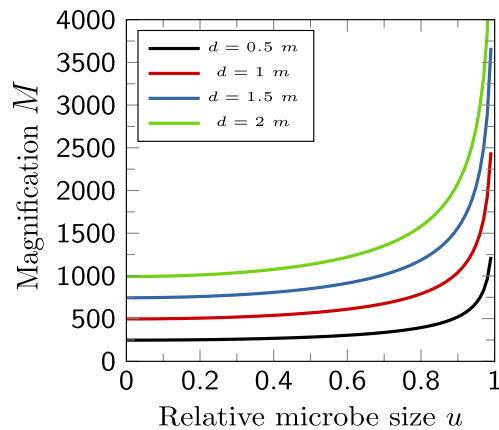


Fig. 3. Magnification M versus the relative microbe size u using different values of d . $n = 1.33$.

with the relative microbe size u till it approaches one where the magnification suddenly diverges. This means that any bacteria with a size close to the water-drop radius ($u \approx 1$) will be blurred. Thus, a water-drop with a very small radius is not recommended to avoid being in the microbe size.

Choosing appropriate distance d , the bacteria images are captured with a camera, which is used to construct the dataset.

2. Data Preparation

The images projected on the screen are collected using a camera. The dataset-building process is done by taking videos of the bacteria shadow floating on the screen and then extracting images from the videos. Moreover, due to the dynamic nature of bacteria moving from the syringe to the water-drop and vice versa, the extracted images can exhibit variations in the bacteria's position, providing a diverse set of images for analysis. This process helps in having a huge amount of images (with a wide range of scenes) on the dataset. After image extraction, the images are separated into the dataset containing *E. coli*, fecal streptococci, both bacteria, and also in the case where there is none of the bacteria. A dataset is then constructed with four classes labeled *E. coli*, fecal streptococci, both, and none. It is important to mention that the class none (no bacteria) is preferably built with certified commercial drinking water. Some image samples contained in the four classes of our dataset are shown in Fig. 4 for rod-like, spherical, both shapes, and none of the shapes representing *E. coli* [Fig. 4(a)], fecal streptococci [Fig. 4(b)], both bacteria [Fig. 4(c)], and none [Fig. 4(d)] of them.

By doing this, with a video of 5 min and 12 s and after extracting noisy images, our dataset contains 5050 useful images that are randomly augmented before being used for classification with CNN.

3. Classification

The diagram representing the architecture of the CNN used can be found in Fig. 5. Since we build our own dataset, it is not recommended to opt for pre-trained CNN such as ResNet-50 as usual [24,25]. Besides, pre-trained CNN uses more layers of convolution and energy-intensive hardware to achieve high classification accuracy. A dense layer is rather employed alongside

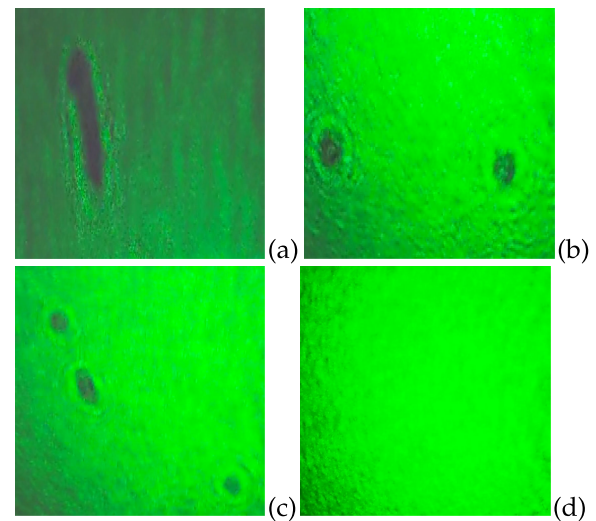


Fig. 4. Different classes of datasets: (a) *E. coli*, (b) fecal streptococci, (c) both (at the corners, spherical for fecal streptococci and in the middle rod-like for *E. coli*), and (d) none.

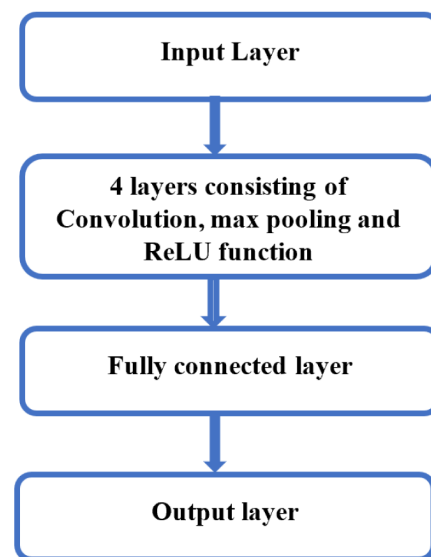


Fig. 5. CNN structure. The parameters are the following: total parameters: 1,109,412 (4.23 MB); trainable parameters: 1,109,412 (4.23 MB); non-trainable parameters: 0 (0.00 B). The number of neurons is 16, 32, 64, and 64 for the four layers, respectively.

a fully connected layer to extract information and organize the data. The outputs are then determined based on the classified results of the dense layer. In our scenario, the input layer is determined by the number of pixels present in the images within the dataset, which is $300 \times 300 \times 3$. Among these images, 4040 images are used for training and 1010 for validation. Besides, to avoid overfitting and ensure a better connection of the CNN and the data, the convolutional layer comprises four subsequent layers, each followed by a maximum pooling layer utilizing the ReLU function. After these layers, a fully connected layer is present.

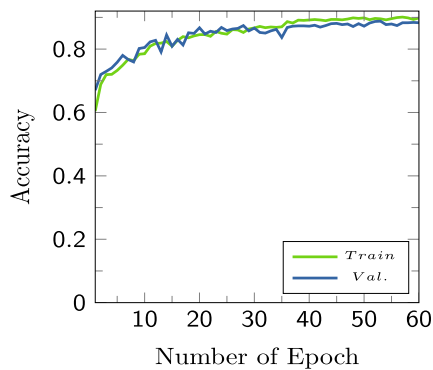


Fig. 6. Training and validation accuracy curves for 60 epochs. Train for training and Val for validation.

To enhance the resilience of our model, we implement online image augmentation, which involves applying random flipping and rotation techniques.

3. RESULTS

This section presents the experimental outcomes of the proposed system for detecting and classifying *E. coli* and fecal streptococci in water. The classification process respecting the architecture given in Fig. 5 provides a validation accuracy as shown in Fig. 6.

Figure 6 illustrates the relationship between the number of epochs and both train and validation accuracies. As the number of epochs increases, the validation accuracy also increases until it almost reaches a plateau during the final 20 epochs, stabilizing at an accuracy of 89% (0.89), indicating a successful classification of the four classes of bacteria in our dataset. This approach used 60 epochs, batch size at 32, and Adam as the optimizer. The learning-rate was set to 0.0001 to achieve high accuracy in extracting invariant features from bacteria images [26]. When the number of epochs is less than 40, the accuracies are unstable, thereby exhibiting fluctuating patterns similar to that of Liedji *et al.* [27] (Fig. 6). These findings demonstrate that increasing the number of epochs enhances validation accuracy and promotes more stable predictions. This behavior resembles that of the pre-trained CNN of [28]. Figure 7 shows both train and validation losses reduced below 0.15 (around 0.13), which is relatively small. The mean square error is also about 0.0415, which indicates the high performance of the model.

4. CONCLUSION

We believe that this work has presented a novel cost-, time-, and equipment-effective system capable of detecting indicator bacteria like *E. coli* and fecal streptococci in water by combining a water droplet method for image acquisition and convolutional neural networks (CNNs) for their classification. Unlike other methods that rely on pre-built datasets, we built our dataset and indicator bacteria classification followed based on the bacteria contour leveraging CNN. We have demonstrated accurate results in all scenarios when detecting and classifying indicator bacteria in water in real-time. Compared to micro-biological and other methods, our approach is faster, cheaper, and easily implementable with available commercial components, and

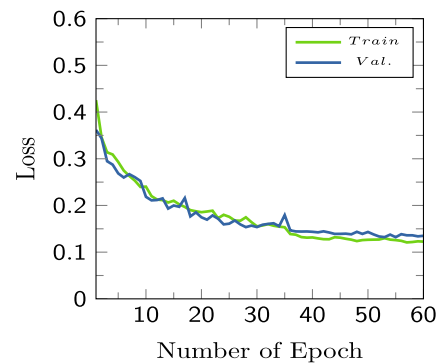


Fig. 7. Training and validation loss curves for 60 epochs. Train for training and Val for validation.

does not require a huge laboratory set. For instance, a computer with Windows 10 Intel Core i5-7300U CPU 2.60 GHz 2.71 GHz, 8 GB RAM with the 64-bit operating system takes approximately a few minutes for the detection, unlike micro-biological methods that take 48–72 h. Moreover, our device presents the advantage of limiting human intervention since there is no sample preparation, which is not without risk of contamination. Combining the optical water droplet method and no pre-trained CNN, we anticipate that our system stands as a rapid and easy water diagnosis technique that could limit the spread of water-borne diseases and facilitate drinking water treatment [29].

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Data availability. Data underlying the results of this paper may be obtained from the authors upon reasonable request.

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